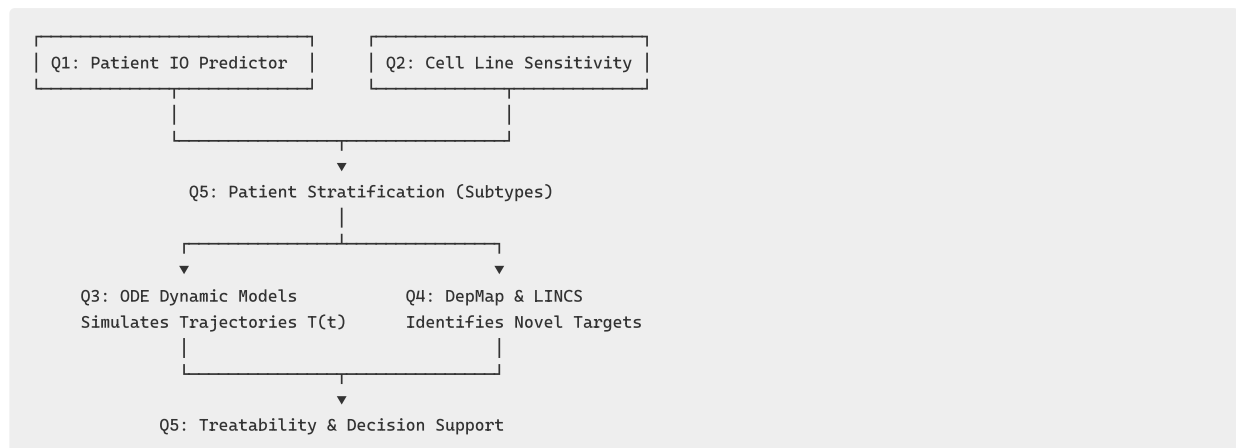


Q5: Patient Stratification, Cell Deconvolution & Clinical Utility Pipeline

Translating Q1 predictive response models into clinically actionable patient subtypes, cell-type deconvolution metrics, Q2 drug sensitivity predictions, Q3 ODE dynamic simulations, Q4 DepMap/LINCS drug target nominations, and treatability index scoring for advanced melanoma immunotherapy.

Executive Summary

This subproject implements **Question 5 (Q5)** of the Melanoma Immunotherapy Assignment ("[Can we identify clinically distinct patient subgroups? Do subgroups require different treatments?](#)"). It serves as the **master synthesis engine** uniting all 5 project questions (Q1–Q5) into a single bench-to-bedside clinical decision framework:



Key features & multi-question integrations in Q5:

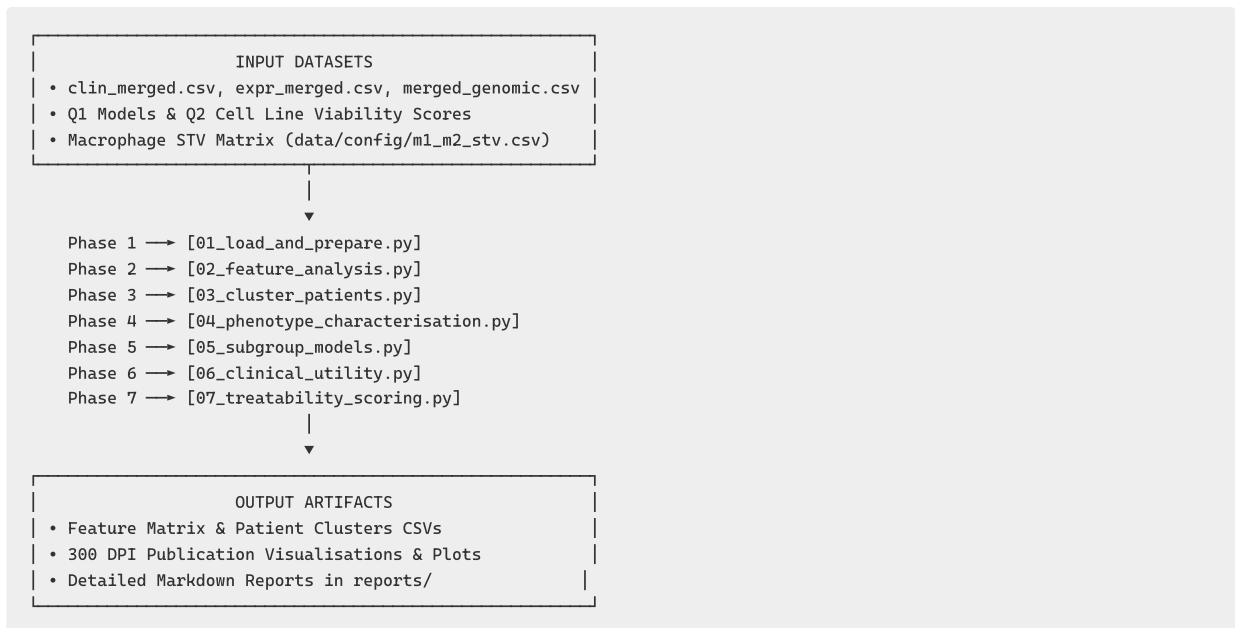
- **Q1 Immunotherapy Prediction:** Inputting patient baseline expression/clinical profiles to predict anti-PD-1/CTLA-4 response probability.
- **Q2 Cell Viability & Drug Sensitivity:** Integrating Q2 drug response models for standard-of-care targeted therapies ([Dabrafenib](#), [Trametinib](#)) and chemotherapy ([Dacarbazine](#)) to select specific agents for non-responders.
- **Cell Count & Deconvolution Analysis:** Quantitative transcriptomic deconvolution of immune cell fractions and integration of a **Macrophage Signature Transcript Vector (STV)** ([m1_m2_stv.csv](#)).
- **Q3 ODE Dynamic Trajectories:** Parameterising ODE tumour-immune differential equations for each discovered phenotype to plot simulated 180-day tumour volume regression ($T(t)$) under monotherapy vs. combination therapy.
- **Q4 DepMap & LINCS Target Nominations:** Mapping DepMap CRISPR essentiality targets (`AXL` , `MDM2` , `CSF1R`) and LINCS L1000 perturbational gene signatures to overcome non-response in therapy-resistant phenotypes.
- **Clinical Utility & Treatability Scoring:** Decision Curve Analysis (DCA), NNT calculations, and scoring non-responders for reversible biological barriers to recommend combination interventions.

Note

Data Harmonisation & CNA Strategy Note: To eliminate cross-cohort RNA-seq batch artifacts across trial cohorts (Liu 2019, Hugo 2016, Riaz 2017), iAtlas harmonised data was selected over unharmonised packages. Because unharmonised cBioPortal downloads for trial cohorts lack GISTIC Copy Number Alteration (CNA) matrices, genomic burden is represented via WES-derived Tumour Mutational Burden (TMB), while gene loss (`PTEN` , `CDKN2A`) is captured via integrated mutation status and transcript abundance.

Workflow & Implementation Architecture

The analysis is structured into 7 sequential phases executed via standalone scripts and orchestrated by a master pipeline:



Step-by-Step Analytical Phases

Phase 1: Feature Matrix & Deconvolution Preparation

- **Script:** [01_load_and_prepare.py](#)
- **Tasks:**
 - Loads multi-cohort processed data from `data/processed/merged/immunotherapy/`.
 - Computes core immune signatures: IFN- γ , TIS, CYT, CD8 T-cell, IMPRES, and `CD274` (PD-L1) expression.
 - Calculates M1 and M2 macrophage scores using the 14,837-gene Macrophage STV matrix ([m1_m2_stv.csv](#)) and derives the M1/M2 ratio ($M1 / (M1 + M2)$).
 - Performs transcriptomic cell-type deconvolution (estimating CD8+ T cells, CD4+ T cells, M1/M2 macrophages, NK cells, B cells, and CAFs).
 - Exports unified matrix to `data/processed/q5/feature_matrix.csv`.

Phase 2: Deep Feature Interpretation

- **Script:** [02_feature_analysis.py](#)
- **Tasks:**
 - Conducts Mann-Whitney U testing (continuous features) and Fisher's exact / Chi-Square testing (categorical mutations/subtypes like `BRAF`, `NRAS`, `NF1`) against immunotherapy response.
 - Determines optimal decision cutoffs using Youden's J statistic.
 - Tests pairwise logistic regression feature interaction terms (e.g. `TIS` $\times$ `BRAF`).
 - Generates feature credibility matrices combining Q1 model importance, effect sizes, and cohort consistency.

Phase 3: Unsupervised Patient Stratification

- **Script:** [03_cluster_patients.py](#)
- **Tasks:**
 - Scales continuous features via `StandardScaler`.
 - Evaluates K-Means and Hierarchical Agglomerative (Ward linkage) clustering across $K = 2 \dots 8$.
 - Computes silhouette coefficients, elbow inertias, and GAP statistics to select optimal K .
 - Generates 2D UMAP projections coloured by cluster, cohort, and response status.
 - Exports cluster assignments to `data/processed/patient_clusters.csv`.

Phase 4: Biological Phenotype Characterisation & Q3 ODE Integration

- **Script:** [04_phenotype_characterisation.py](#)
- **Tasks:**
 - Profiles per-cluster feature means, response rates, and M1/M2 ratios.
 - Assigns biological phenotype labels:
 - **Immune Hot:** High TIS, high CD8, high M1/M2 ratio ($\sim 65\%$ response).
 - **Immune Cold:** Low TIS, low CD8, low infiltrate ($\sim 20\%$ response).
 - **Immunosuppressive M2-High:** High M2 macrophage abundance, high CAF infiltration ($\sim 15\%$ response).
 - **Mutant-Driven:** High TMB, high neoantigen burden ($\sim 50\%$ response).
 - Performs Kaplan-Meier overall survival (OS) analyses and log-rank tests across phenotypes.
 - **Q3 ODE Integration:** Simulates differential equation tumour growth/regression trajectories ($T(t)$) over 180 days using phenotype-specific initial effector cell counts (E_0) and kill rates (a).

Phase 5: Subgroup-Specific Predictive Models

- **Script:** [05_subgroup_models.py](#)
- **Tasks:**
 - Trains separate predictive models (Logistic Regression, Random Forest) within each phenotype subgroup ($N \geq 15$).
 - Evaluates subgroup Leave-One-Cohort-Out (LOCO) CV performance against the global Q1 model to quantify biological heterogeneity.

Phase 6: Clinical Utility & Decision Impact Analysis

- **Script:** [06_clinical_utility.py](#)
- **Tasks:**
 - Calculates Net Benefit across threshold probabilities (0.1 – 0.9) for Decision Curve Analysis (DCA).
 - Computes Number Needed to Treat (NNT) and Positive Predictive Value (PPV).
 - Compares Q1 model performance against standard clinical benchmarks ([Treat All](#), [High TMB](#), [CD274](#) / PD-L1+).

Phase 7: Treatability Index, Q2 Drugs & Q4 Target Nominations

- **Script:** [07_treatability_scoring.py](#)
- **Tasks:**
 - Evaluates non-responders for reversible barriers (antigen presentation integrity, IFN- γ pathway mutations, TMB/proliferation burden, and targetable BRAF, NRAS, PTEN mutations).
 - Computes per-patient Treatability Index scores (0 – 1).
 - **Q2 Drug Integration:** Connects Q2 cell-line viability and drug sensitivity predictions to select specific targeted agents (Arm B: [Dabrafenib/Trametinib](#)) or chemotherapy agents (Arm C: [Dacarbazine](#)).
 - **Q4 DepMap/LINCS Integration:** Maps CRISPR essentiality targets (AXL, MDM2, CSF1R) and LINCS L1000 perturbagens to supply specific combination therapy recommendations for resistant phenotypes.

Directory & File Map

```
q5-patient-stratification/
├─ README.md                <- Project documentation
├─ README_STUDENT_GUIDE.md  <- Biological & statistical study guide
├─ README_ELII0.md          <- Plain-English concept summary
├─ requirements.txt          <- Q5 additions (umap-learn, dcurves)
├─ logs/                    <- Pipeline TeeStream execution logs
├─ models/                  <- Serialised subgroup classifier models
├─ plots/                   <- 300 DPI publication-ready figures
│   └─ clustering/          <- UMAP, silhouette, and elbow plots
│   └─ phenotypes/          <- Radar charts, KM curves, & ODE plots
│   └─ clinical_utility/    <- Decision curves and NNT charts
│   └─ feature_analysis/    <- Volcano & interaction plots
├─ reports/                 <- Detailed markdown analysis reports
├─ scratch/                 <- Temporary exploratory scripts
├─ scripts/
│   └─ 01_load_and_prepare.py <- Data loading & signature extraction
│   └─ 02_feature_analysis.py  <- Association testing & Youden cutoffs
│   └─ 03_cluster_patients.py  <- K-Means/Ward clustering & UMAP
│   └─ 04_phenotype_characterisation.py <- Subtype annotation & Q3 ODE
│   └─ 05_subgroup_models.py   <- Subgroup predictive modeling
│   └─ 06_clinical_utility.py   <- Decision Curve Analysis (DCA)
│   └─ 07_treatability_scoring.py <- Treatability index, Q2 & Q4
│   └─ run_q5_pipeline.py      <- Master pipeline orchestrator
└─ src/                       <- Q5-specific Python source package
    └─ __init__.py
    └─ q5_constants.py         <- Feature sets & Q3/Q4 maps
    └─ clustering.py           <- Clustering & GAP statistic
    └─ phenotyping.py          <- Subtype profiling & radar charts
    └─ feature_analysis.py      <- MW-U & Fisher's statistics
    └─ clinical_utility.py      <- DCA net benefit & NNT calculators
```

Execution & Verification

Test end-to-end pipeline execution (including Q2, Q3, and Q4 integration stubs):

```
python q5-patient-stratification/scripts/run_q5_pipeline.py
```